

Causal Relationships and Programming Outcomes:

A Transcranial Magnetic Stimulation Experiment

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Introduction

- Prior brain imaging studies have linked spatial reasoning and programming, but only *observationally*.
- We investigated whether this link is *causal* in a controlled human experiment.

Spatial Reasoning: The cognitive skill of mentally manipulating and reasoning about objects.

Transcranial Magnetic Stimulation (TMS): A non-invasive neurostimulation technique that temporarily disrupts activity in a targeted brain region.

- Applying this *intervention* enables controlled study of cognitive mechanisms.
- Used to study: working memory, creativity, mental math, human language processing.

Motivation

- Spatial reasoning is well-studied in psychology and is highly malleable.
- If a causal relationship exists, known training methods could *transfer* to CS domains.
- If the relationship is non-causal, training could detract from student learning.
- A causal relationship could help explain disparities in CS outcomes.

Neurostimulation of brain areas used for spatial reasoning affects programming outcomes.

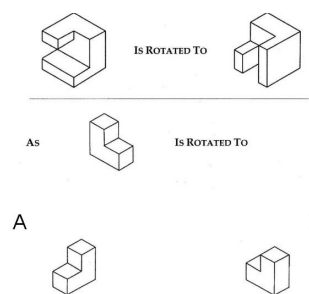
Experiment

- 16 participants $\times$ 3 interventions (brain regions):
 - *Supplementary Motor Area (SMA)* — causal link w/ SR
 - *Primary Motor Cortex (M1)* — observational link w/ SR
 - *Cranial Vertex* — active control, no link
- 63 multiple-choice tasks (across 5 topic areas) $\times$ 3 forms
- Preregistered, single-blind, and randomized
- Metrics: **response time**, **accuracy**, and **self-reporting**



Left — TMS session. Researcher (right) uses per-participant localization (left screen) and hand-held magnetic coil (center) on participant's scalp (seated) to induce current in a brain region.

Right — Example spatial reasoning stimulus.



Given the top array, after performing the first bubble in bubble sort, which candidate array will be the result?

indices	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
nums	78	9	53	21	11	63	98	1	82	39	90	54	68	15	13

A:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
9	78	53	21	11	63	98	1	82	39	90	54	68	15	13

B:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
9	53	78	21	11	63	98	1	82	39	90	54	68	15	13

Example programming task. Also included were linked lists, trees, code comprehension, and Big- O code complexity.

Data Analysis

- Regression-based approach
- Predict a metric, e.g. response time, from brain region stimulated and control variables
- Model systematic ("fixed") and heterogeneous ("random") effects

```
log(response_time)
~ domain  $\times$  session_num
+ (1|id) + (1|stimulus)
+ (1|id : domain) + (1|id : condition)
```

		Std.	2.5%	97.5%
Random Effect	Vari.	Dev.	C.I.	C.I.
Task	.204	.452	.398	.517
Participant by Task Type	.019	.137	.100	.190
Participant by Brain Region	.010	.099	.066	.143
Participant	.037	.193	.118	.308
Residual	.175	.418	.404	.433

Model parameters predicting log-transformed response times. Point estimates and 95% confidence intervals.

References

- [1] Illinois Geometry Lab. IGL Poster Template. *University of Illinois at Urbana-Champaign Department of Mathematics*, 2017.



← See the paper!

